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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 03/20/2006 |
| **PI Name:** | Bolander, Mark E M.D. | | |
| **Project Title:** | The Effect of Microgravity on Fracture Healing/Ultrasound as a Possible Countermeasure | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Bone Loss Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 55905 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | NSBRI |
| **Start Date:** | 04/01/2001 | **End Date:** | 05/31/2004 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | N/A | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | BL.002.07 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | The NSBRI Conference that was convened in Clear Lake, Texas, on November 16-17, 2000, identified fracture healing during spaceflight as an area where further information would be required to appropriately prepare for long-term space missions, and developing countermeasures to restore normal fracture healing was identified as a priority for current research. The RFA dated February 22, 2000, (NSBRI II 00-01) requested studies evaluating the effect of spaceflight on fracture healing and developing countermeasures. This application is submitted in response to that RFA. Our current understanding of bone physiology suggests that fracture healing will be abnormal in the microgravity environment. This hypothesis is supported by two published studies, the first an abstract reporting abnormal healing in rats undergoing hindlimb unloading (27), the second a manuscript (in Russian) that describes abnormal fracture healing in five rats with fibula fractures flown on Cosmos-2044 (28). This latter study reports that abnormalities seen in fracture healing after spaceflight were duplicated in the hindlimb unloading model. The goals of the experiments proposed in this application | | |

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| | are 1) to confirm the previous reports that microgravity adversely affects fracture healing, and 2) to determine if ultrasound treatment, which has been shown to accelerate fracture healing in clinical studies, will reverse the impaired cellular events in fracture healing that are related to microgravity. If ultrasound does not act as an effective countermeasure we will undertake detailed evaluation of our histologic samples to identify potential targets for other countermeasures. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | x |
| **Task Progress:** | No progress reported. |
| **Bibliography Type:** | Description: (Last Updated: 02/27/2004) |